

Cost-Effective Colleges and Intergenerational Mobility: Evidence from the U.S. College Scorecard

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Github Repository: <https://github.com/jethrochen0709/Econ570FinalProject>

1. Introduction

In this project, we examine how college cost, student socioeconomic background, and institutional characteristics shape long-run economic outcomes for U.S. undergraduates. We address four key questions: (1) how net price relates to median earnings ten years after entry across different institutional types; (2) whether net price remains predictive of earnings after accounting for family income as a key confounder; (3) which institutions both enroll large shares of low-income students and over-perform on earnings and repayment; and (4) which colleges best support upward intergenerational mobility. Together, these analyses provide a view of how cost, access, and institutional effectiveness interact to shape student outcomes.

Overall, we find that while higher-priced colleges tend to report higher earnings ten years after entry, much of this relationship reflects differences in students' family income and academic preparation. At the same time, many public and selected specialized institutions deliver strong earnings and mobility outcomes at relatively low cost.

2. Analysis

2.1 Data

Using the U.S. College Scorecard from the CMU S&DS Data Repository, we analyzed bachelor's-degree institutions to assess how institutional cost, sector, and student background influence long-run outcomes. We focus on median earnings ten years after entry, with three-year loan repayment rates and 10-year monthly loan payments capturing debt burden, and use net price as our cost measure. Student background is represented by constructed measures of family income at entry and academic ability from SAT/ACT scores, allowing us to evaluate how cost, access, and institutional characteristics relate to students' economic returns.

2.2 Cost and Earnings

College cost is often treated as a signal of institutional quality, but the relationship between net price and later earnings is imprecise. **Figure 1** shows a modest upward trend between average net price and median earnings ten years after entry, but with substantial dispersion. Many low-cost public institutions achieve earnings comparable to far more expensive private colleges, while some high-cost institutions produce only moderate outcomes, indicating that net price alone is a noisy indicator of institutional value.



Figure 1: Net Price vs Earnings by Institution Type

Table 1 summarizes sector averages across institution types. Public colleges have the lowest mean net price while achieving earnings comparable to private nonprofits. Private nonprofits charge more but deliver only modestly higher earnings, whereas for-profit colleges are both the most expensive and the lowest earning on average. These patterns reinforce **Figure 1**, showing that public institutions often match private nonprofits at far lower cost, while for-profits underperform on both dimensions.

Institution Type	Mean Net Price	Mean Median Earnings After 10 Years
Private for-profit	\$25,615.82	\$48,083.58
Private nonprofit	\$22,696.70	\$53,068.45
Public	\$13,882.23	\$51,853.96

Table 1: Mean Net Price and Mean Median Earnings after 10 Years by Institution Type

Since sector averages can mask important differences within each group, we also examine cost-effectiveness at the institutional level. **Table 2** reports the top 10 institutions ranked by our earnings-per-dollar index, which measures median earnings per dollar of net price. Most of the top performers are public universities, with MIT appearing as a single notable private college as an exception, reinforcing that many public institutions provide exceptional value and that high prices do not guarantee high returns.

$$EarningsPerDollar_i = Median\ Earnings_i / Net\ Price_i$$

Equation 1: Earnings-per-Dollar Index

Institution Name	State	Institution Type	Net Price	Median Earnings after 10 Years	Earnings per Dollar Spent
CUNY Bernard M Baruch College	NY	Public	\$1,936	\$71,078	\$36.71
Luther Rice College & Seminary	GA	Private nonprofit	\$1,392	\$49,700	\$35.70
CUNY City College	NY	Public	\$2,061	\$58,052	\$28.17
CUNY Hunter College	NY	Public	\$2,124	\$58,822	\$27.69
California State University–Los Angeles	CA	Public	\$2,037	\$54,219	\$26.62
California State University–San Bernardino	CA	Public	\$2,232	\$55,258	\$24.76
Massachusetts Institute of Technology	MA	Private nonprofit	\$5,084	\$124,213	\$24.43
CUNY Brooklyn College	NY	Public	\$2,393	\$57,665	\$24.10
CUNY John Jay College of Criminal Justice	NY	Public	\$2,254	\$54,207	\$24.05
CUNY Lehman College	NY	Public	\$2,615	\$52,882	\$20.22

Table 2: Top 10 Universities by Earning per Dollar Spent

2.3 Family Income as a Confounder

The dispersion in **Figure 1** indicates that institutional type alone cannot explain why colleges with similar prices produce very different earnings. A key part of the explanation lies in student background. Family income strongly shapes both college choice and later economic outcomes, making it an important confounder in the cost–earnings relationship. As shown in **Figure 2**, family income is moderately correlated with net price ($r = 0.517$), indicating that higher-income students tend to enroll at more expensive institutions. **Figure 3** shows a similarly strong positive association between family income and earnings, suggesting that students from wealthier families earn more regardless of where they enroll. Together, these patterns imply that part of the observed cost–earnings gradient reflects differences in who attends colleges rather than differences in institutional quality.

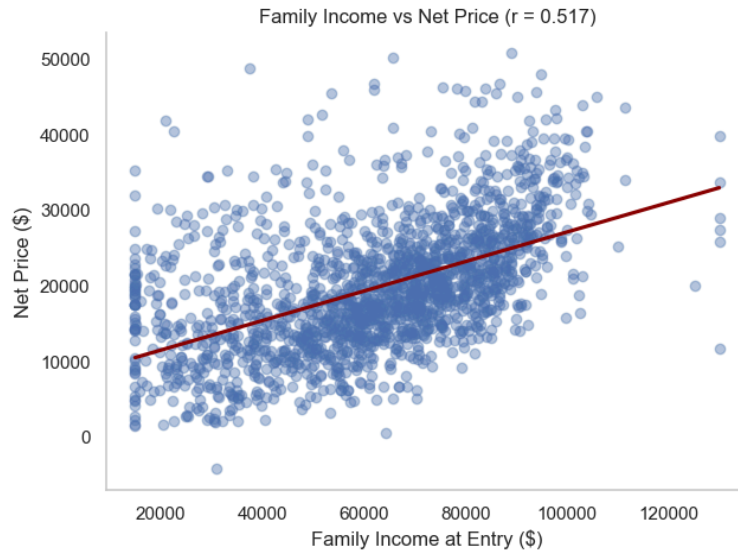


Figure 2: Family Income vs Net Price

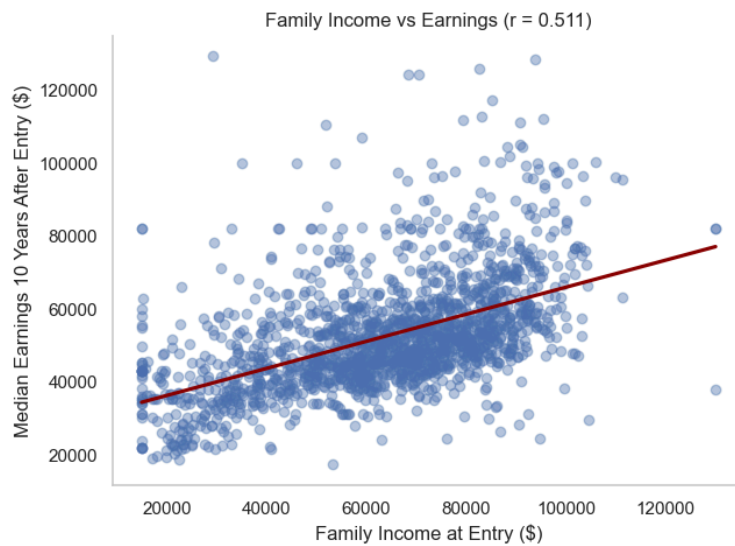


Figure 3: Family Income vs Earnings

To quantify this confounding, we estimate two OLS regressions, summarized in **Table 3**. The simple model in **Equation 2** relates median earnings only to net price and finds a positive association. The full model in **Equation 3** adds family income, test scores, institution type, and log enrollment. Once these controls are included, the net-price coefficient becomes much smaller, indicating that much of the apparent payoff to higher net price is associated with differences in student background rather than institutional effectiveness. In the full model, family income and test scores are strongly associated with earnings, and private nonprofits retain a modest advantage over publics, while for-profits show no clear benefit. Once family income and student ability are accounted for, the relationship between net price and earnings becomes

much weaker, reinforcing that institutional price alone is not a reliable indicator of return on investment.

$$\text{Median Earnings}_i = \beta_0 + \beta_1 \text{Net Price}_i + \varepsilon_i$$

Equation 2: Simple Net Price–Earnings OLS Model

$$\text{Median Earnings}_i = \beta_0 + \beta_1 \text{Net Price}_i + \beta_2 \text{Family Income}_i + \beta_3 \text{Test Index}_i + \beta_4 \text{Institution Type}_i + \beta_5 \log(\text{Enrollment}_i) + \varepsilon_i$$

Equation 3: Full Earnings Determinants OLS Model

Variable	Simple	Full
Net price	0.845144	0.319137
Family income		0.019832
Test index		9263.892768
Private nonprofit		-3235.80555
For-profit		

Table 3: OLS Regression Results Comparison

2.4 Low-Income Access & Value-Added Earnings Model

This section evaluates how effectively institutions support economic outcomes for students from low-income backgrounds by focusing on two factors: access, measured by median family income and Pell Grant share, and outcomes, measured by median earnings ten years after entry and the three-year repayment rate, which reflects the share of federal student loan borrowers actively reducing their loan principal three years into repayment. Together, these indicators identify institutions that enroll large numbers of low-income students while delivering strong long-run economic outcomes.

2.4.1 Institutions Serving Low-Income Students with Strong Outcomes

Using the top quartile of Pell Grant share and the bottom quartile of family income as selection criteria, we identify a group of institutions that disproportionately serve low-income students. As shown in **Table 4**, these colleges enroll students from families with an average income of \$31,269 and a median income of \$32,211, alongside a high average Pell Grant share of 63.65%. Despite serving largely disadvantaged populations, these institutions maintain moderate net prices, averaging \$14,942, making them substantially more affordable than many four-year programs nationwide.

When focusing on outcomes, students attending these institutions earn an average of **\$39,957** ten years after entry and a median of **\$38,092**. While these numbers are lower than the national average for all 4-year institutions at **\$52,838**, they still represent meaningful economic gains given the students' backgrounds and low cost of attendance. Repayment rates at these institutions average at **38%** which indicates that a good portion of borrowers are successfully reducing their loan balances.

Overall, Table 4 highlights that high-access, low-income serving institutions offer relatively affordable educational opportunities and still help students earn a modest economic outcome after 10 years.

Variable	Mean	Median
Family Income	\$31,268.83	\$32,211.44
Pell Grant Share	63.65%	62.51%
Net Price	\$14,942.03	\$14,211.00
Median Earnings After 10 Years	\$39,956.63	\$38,092.00
3 Year Repayment Rate	38.06%	36.16%

Table 4: Summary of Low-Income–Serving Institutions

2.4.2 Institutions with Over-performance

In this part, we examine which institutions “over-perform” on earnings after accounting for student background. Using the value-added model in **Equation 4**, we regress median earnings ten years after entry on family income, test index, institution type, and log undergraduate enrollment. The residual from this regression, standardized as a z-score, serves as our value-added measure, where higher values indicate earnings above what would be expected given students' backgrounds and institutional characteristics.

$$Earnings_i = \beta_0 + \beta_1 FamilyIncome_i + \beta_2 TestIndex_i + \beta_3 InstitutionType_i + \beta_4 \log(Enrollment_i) + \varepsilon_i$$

Equation 4: Value-Added OLS Regression Model

Using the OLS model in **Equation 4**, we assess how effectively institutions raise earnings beyond what would be expected given student backgrounds. The distribution of value-added scores in **Figure 4** shows meaningful variation exists across sectors. Private nonprofit institutions comprise most of the sample and exhibit a relatively wide distribution (standard deviation 1.116445), with a mean effectively equal to zero (5.36E-12), indicating that earnings generally align with predictions on average (**Table 5**). Public institutions display a tighter distribution (standard deviation 0.753339), suggesting more consistent but less extreme deviations from predicted earnings. Private for-profit institutions appear only minimally due to limited data; the two included schools have a high standard deviation (3.482791) but an average value-added score near zero.

Overall, the value-added results demonstrate that while many institutions cluster near their expected performance, there are still a subset of primarily private nonprofit schools that substantially outperform predictions. This indicates that institutional factors beyond student background may meaningfully influence long-run earnings.

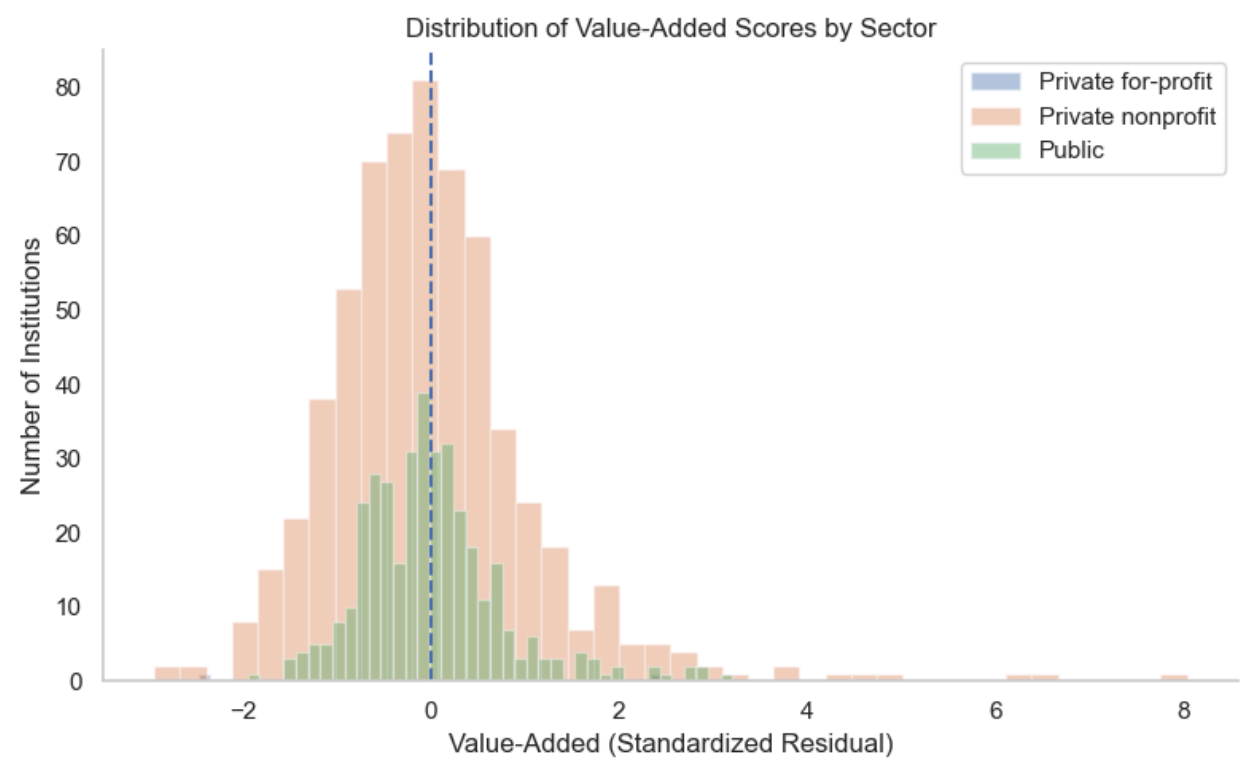


Figure 4: Distribution of Value-Added Earnings Scores by Institutional Type

Institutional Type	Count	Mean	Median	Standard Deviation
Private For-Profit	2	1.41E-11	1.41E-11	3.482791
Private Nonprofit	615	5.36E-12	-0.1018412	1.116445
Public	372	-8.94E-12	-0.07562514	0.753339

Table 5: Value-Added Earnings Scores by Institutional Type

2.5 Intergenerational Mobility

This final section evaluates how institutions support upward economic mobility, particularly for low-income students, by constructing a composite mobility score that captures affordability, student background, and post-college outcomes.

This mobility composite aggregates percentile ranks (0–1) on net price, median family income, median earnings ten years after entry, and the three-year repayment rate, so that higher scores correspond to lower net price and family income but higher earnings and repayment:

$$MobilityScore_i = (1 - NetPriceRank_i) + (1 - FamilyIncomeRank_i) + EarningsRank_i + RepaymentRank_i$$

Equation 5: Mobility Composite Ranking

Unlike **Section 2.4.1**, which restricts attention to clearly low-income-serving colleges (high Pell Grant share and low family income), this mobility composite is applied to all institutions and summarizes outcomes conditional on cost and student family income. **Table 6** shows that the top colleges on this metric include a mix of elite private universities (e.g. Princeton University, Stanford University, Pomona College, and MIT) and public institutions like (e.g. UC–Irvine, several University of Connecticut campuses, CUNY Bernard M Baruch College, and the University of Washington–Tacoma Campus). This mix indicates that high mobility value can emerge from multiple institutional types. Overall, the mobility ranking highlights institutions that balance affordability, access, and long-run economic returns.

Institution Name	Net Price	Median Family Income	Median Earnings After 10 Years	3 Year Repayment Rate	Mobility Score
Princeton University	\$11,080.00	\$51,911.89	\$110,433.00	86.3946%	3.59228
Stanford University	\$7,200.00	\$59,171.28	\$106,987.00	89.92%	3.57117
Pomona College	\$9,170.00	\$54,752.58	\$74,305.00	85.1064%	3.509349
University of California-Irvine	\$9,691.00	\$46,988.13	\$76,593.00	77.5523%	3.501206
University of Connecticut-Waterbury Campus	\$7,599.00	\$50,243.14	\$76,076.00	77.9809%	3.496984
University of Connecticut-Stamford	\$10,225.00	\$53,460.00	\$76,076.00	77.9809%	3.418577
University of Connecticut-Hartford Campus	\$9,370.00	\$54,578.24	\$76,076.00	77.9809%	3.417973
CUNY Bernard M Baruch College	\$1,936.00	\$30,776.36	\$71,078.00	65.1617%	3.408926
University of Washington-Tacoma Campus	\$8,541.00	\$56,461.54	\$74,063.00	79.0376%	3.405308

Massachusetts Institute of Technology	\$5,084.00	\$70,468.01	\$124,213.00	91.8228%	3.398673
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Table 6: Mobility Composite Rankings

3. Conclusion

Our analysis shows that higher-priced colleges generally report higher median earnings 10 years after entry, but this relationship is noisy and strongly sector-specific. Public institutions often achieve earnings comparable to private nonprofits at much lower net prices, while for-profit colleges tend to underperform on both cost and outcomes. Family income is closely related to both net price and later earnings, and once family income and test scores are included in OLS regressions, the independent association between net price and earnings becomes substantially weaker.

Examining low-income access and institutional performance further clarifies how colleges contribute to intergenerational mobility. Some institutions that serve large shares of low-income students—measured by high Pell Grant participation and low median family income—provide affordable pathways with modest earnings and repayment outcomes. Other institutions over-perform relative to expectations after accounting for student background, suggesting that institutional practices can meaningfully influence long-run earnings. When affordability, student background, earnings, and repayment are combined into a single mobility measure (shown in Table 6), both elite private universities and accessible public campuses rank highly. Overall, these findings show that no single institutional type dominates; instead, colleges promote economic mobility through different combinations of access, affordability, and long-run outcomes.